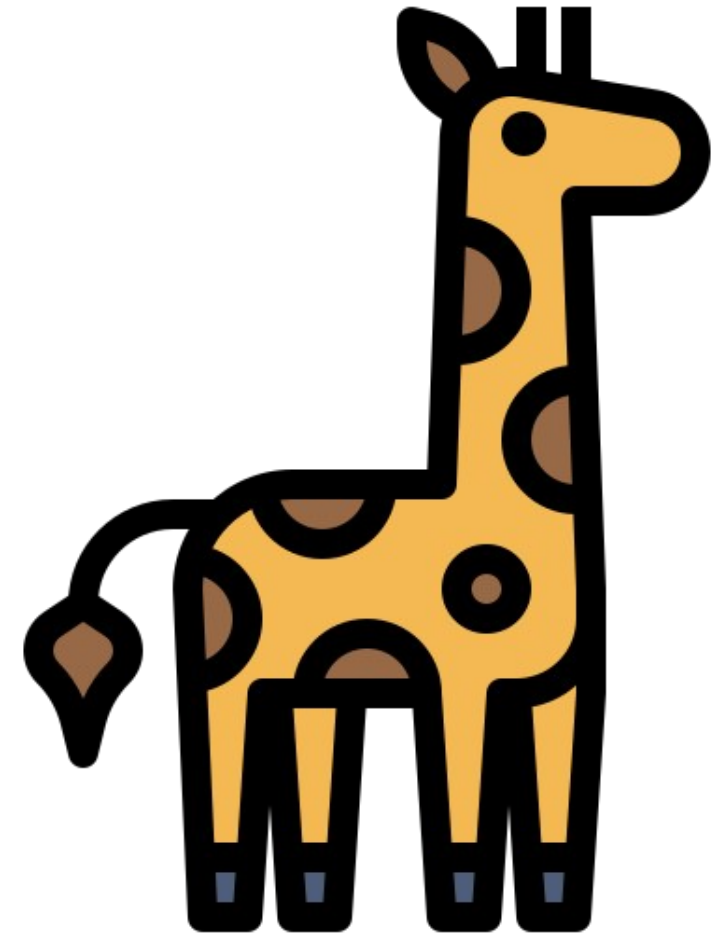




Hi!

It's truly great to see you all again



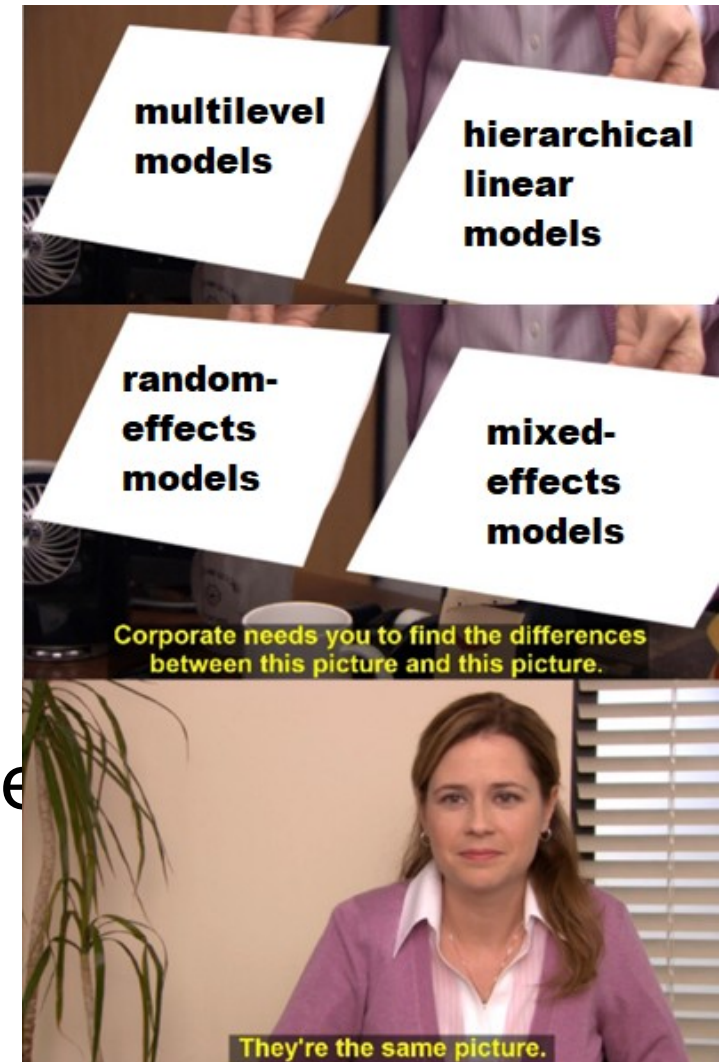
My thoughts about classes this semester

- Time for getting help with assignments
- Short presentations with my perspectives/explanations of stuff
- Sometimes extra exercises or extra explanations
- Let me know if you have wishes!! (both for format and content)

Plan for the class

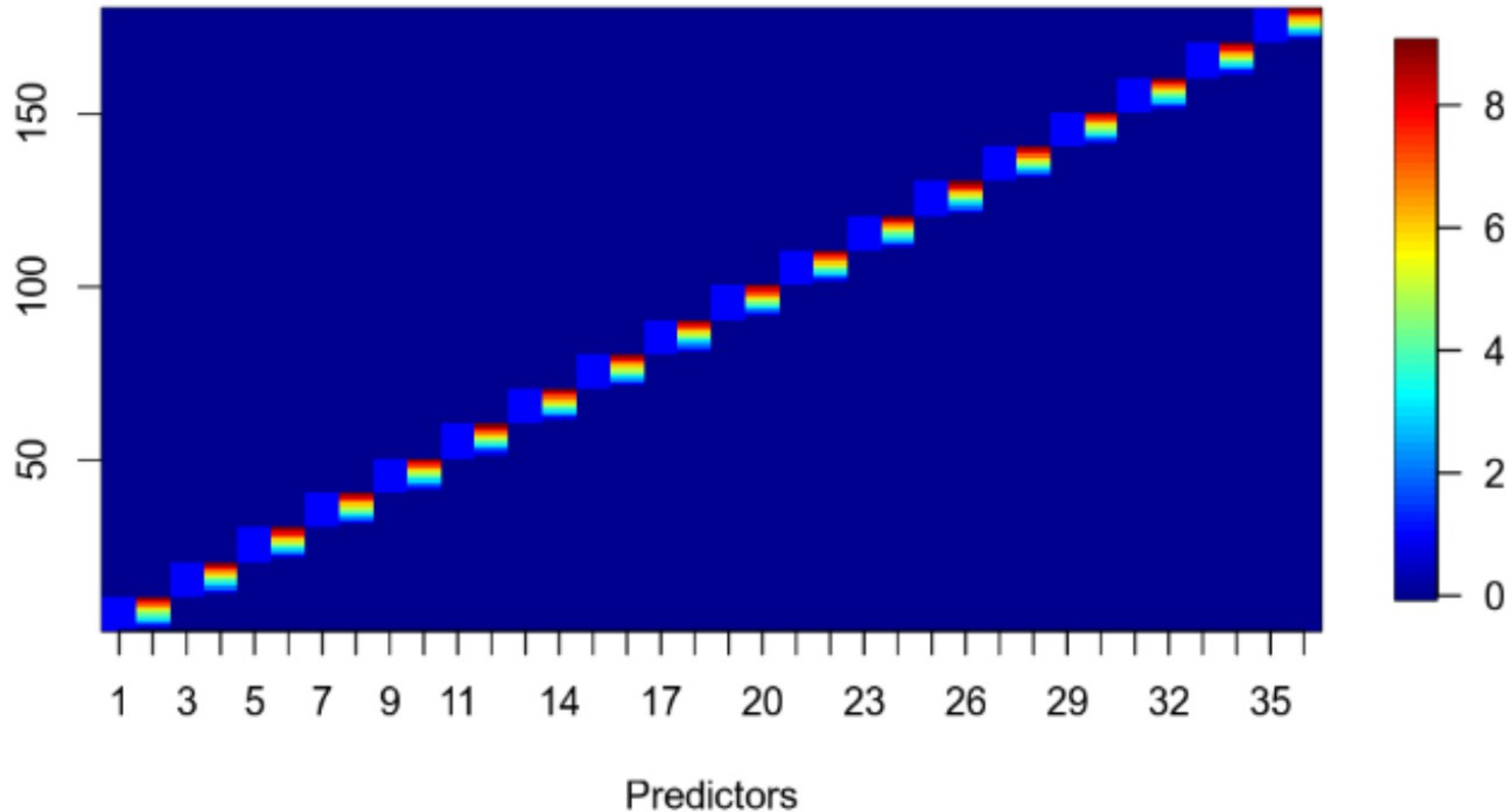
- Understanding the Z matrix
- And the variance-covariance matrix
- How-to in R
- A little bit about pooling

Time to work on Assignment 1 or exercise



Return of the design matrix

Design matrix: Z



The math is the same as last semester, so we just have to take it easy and keep calm

Notation clarification

$$Y = \boxed{X\beta} + \boxed{Zb} + \epsilon$$

Level 1 Level 2

Fixed effects Random effects

$$b \sim N(0, \hat{\Sigma})$$

Difference between level 1 and level 2:
We model the values in the b-vector to be normally distributed (because they are related).

This is a matrix, not a sum!

Notation clarification

$$b \sim N(0, \hat{\Sigma})$$

Random slope

Random intercept

$$\hat{\Sigma} =$$

Covariance of intercept and slope coefficients	Variance of intercept coefficients
Variance of slope coefficients	Covariance of intercept and slope coefficients

Random intercept

Random slope

Does our model give subjects similar or different intercepts? (same for slopes)

Do subjects with higher intercepts also have higher slopes? Or are they uncorrelated?

What does this look like in R?

```
`{r}`  
mod3 = lmer(data = df, Reaction ~ Sleep_deprivation + (Sleep_deprivation|Subject))  
  
mod3  
`{r}`
```

Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Sleep_deprivation + (Sleep_deprivation | Subject)
Data: df
REML criterion at convergence: 1404.094
Random effects:

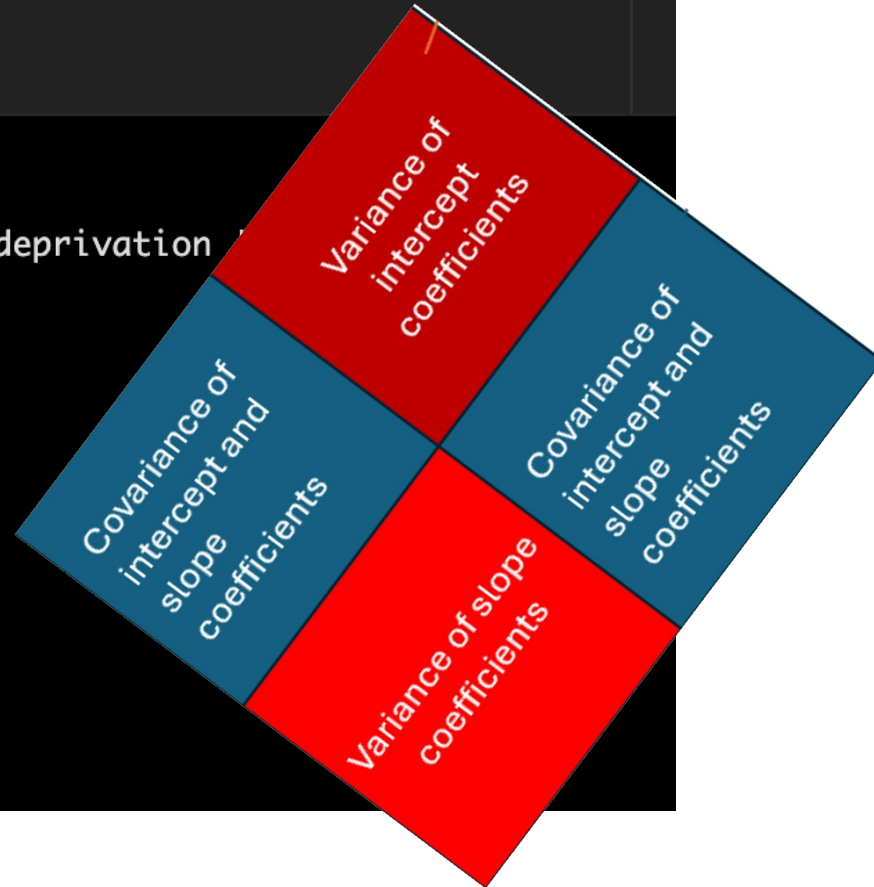
Groups	Name	Std.Dev.	Corr
Subject	(Intercept)	30.957	
	Sleep_deprivation	6.766	0.18

Residual 25.526

Number of obs: 144, groups: Subject, 18

Fixed Effects:

(Intercept)	Sleep_deprivation
267.97	11.44



General how-to in R

Random intercept:

```
Model <- lmer(reaction ~ sleep_deprivation + (1 |Subject), data=df)
```

Random slope only model:

```
Model <- lmer(reaction ~ sleep_deprivation + (0+sleep_deprivation |Subject),  
data=df)
```

Random slope and Intercept:

```
Model <- lmer(reaction ~ sleep_deprivation + (sleep_deprivation|Subject ), data=df)
```

Using lmer() automatically uses pooling. For non-pooled intercept use:

```
lm(reaction ~ sleep deprivation + Subject, data = df)
```

For non-pooled intercept and slope use:

```
lm(reaction ~ sleep deprivation * Subject, data = df)
```


The big scary fraction

$$\alpha_j = \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{X}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha$$

exact anyway, so don't worry too much about it

away's:

the less data we have for someone, the more we pool (use data from other subjects)

the more all-over-the-place (varied) someone's data is, the more we pool (because variance = uncertainty)